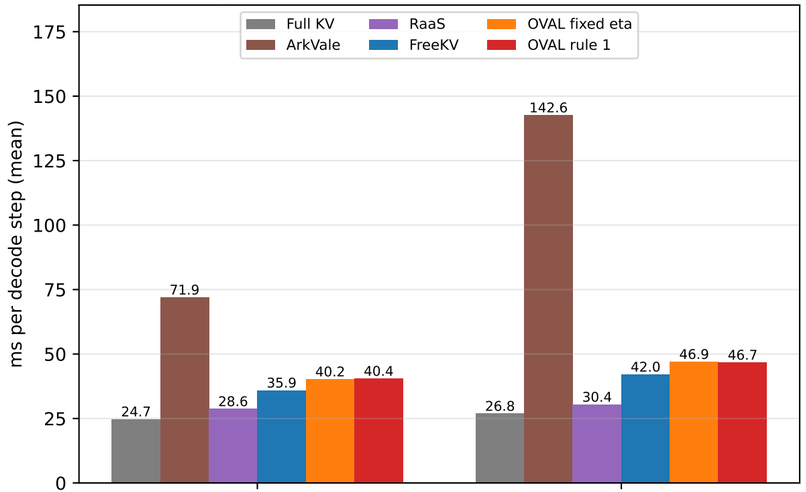
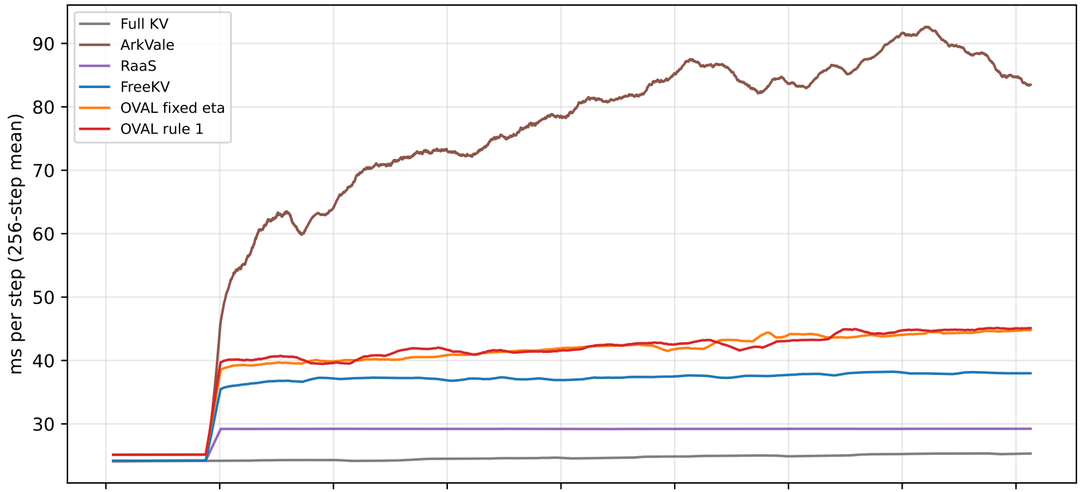
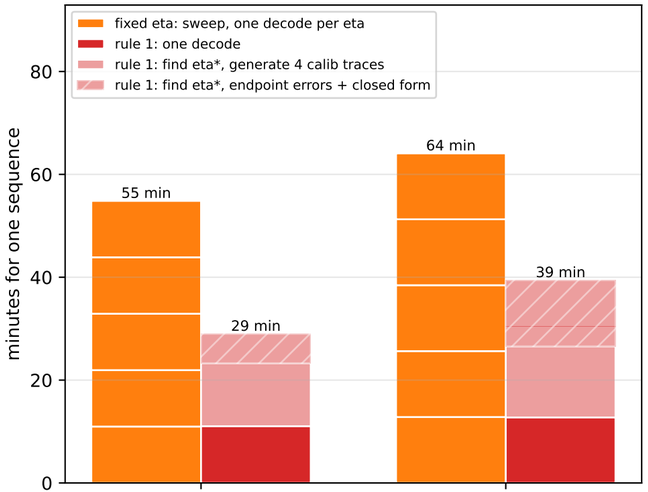

Decode latency, AIME24 problem 0, 16384 decoded tokens, batch 1
budget 2048 tokens (incl. 512 sink + 512 recent), A6000
Full KV
ArkVale
RaaS
FreeKV
OVAL fixed eta
OVAL rule 1
ms per decode step (mean)
0
25
50
75
100
125
150
175
24.7
71.9
28.6
35.9
40.2
40.4
26.8
142.6
30.4
42.0
46.9
46.7
R1-Distill-Qwen-7B
R1-Distill-Llama-8B


R1-Distill-Qwen-7B: per-step decode time over the generation
Full KV
ArkVale
RaaS
FreeKV
OVAL fixed eta
OVAL rule 1
ms per step (256-step mean)
30
40
50
60
70
80
90
0
2000
4000
6000
8000
10000
12000
14000
16000
decode step


Setting eta for one 16384-token sequence
eta* is found once per (model, task), reused after
fixed eta: sweep, one decode per eta
rule 1: one decode
rule 1: find eta*, generate 4 calib traces
rule 1: find eta*, endpoint errors + closed form
minutes for one sequence
0
20
40
60
80
55 min
29 min
64 min
39 min
R1-Distill-Qwen-7B
R1-Distill-Llama-8B